

Chapter 7: Mole concept and stoichiometry

Relative Atomic Mass (A_r)

The relative atomic mass (A_r) of an element is the average mass of one atom of that element relative to $\frac{1}{12}$ the mass of an atom of carbon-12.

- no unit

Relative Molecular Mass (M_r)

The relative molecular mass of a molecular substance is the average mass of one molecule of that substance relative to $\frac{1}{12}$ the mass of an atom of carbon-12.

- no unit

Relative Formula Mass (M_r)

The relative formula mass of an ionic compound is the average mass of one unit of that compound relative to $\frac{1}{12}$ the mass of an atom of carbon-12.

- no unit

Percentage Mass

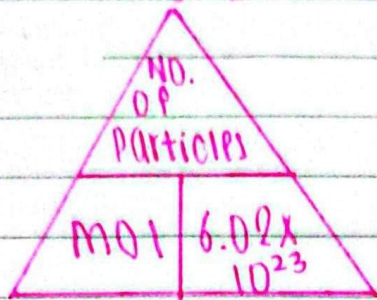
$$\text{Percentage Mass} = \frac{\text{no. of atoms of the element in formula} \times A_r \text{ of element}}{M_r \text{ of compound}} \times 100\%$$

Mass of an element in a sample of compound

$$\text{Mass of element} = \frac{\text{no. of atoms of the element in formula} \times A_r \text{ of element}}{M_r \text{ of compound}} \times \text{sample mass}$$

The Avogadro constant and the mole

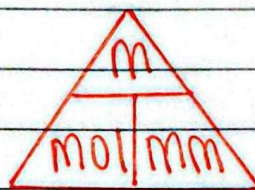
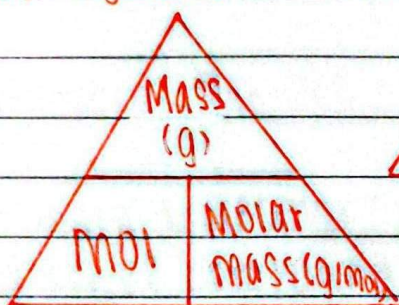
1 mol of any substance always contains 6.02×10^{23} particles.



Molar mass

Molar mass is the mass equal to A_r or M_r .

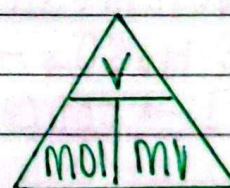
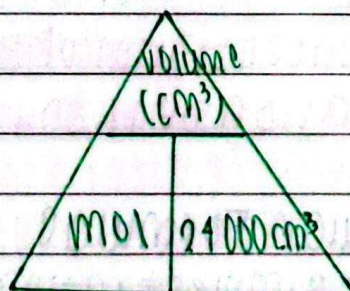
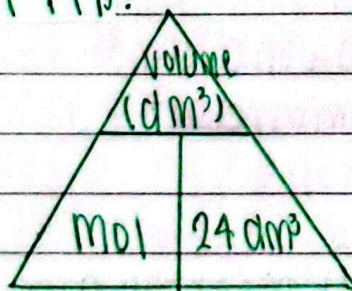
Unit $\rightarrow$ g/mol



This formula applies to all solid, liquid, gas

Avogadro's Law and Molar volume

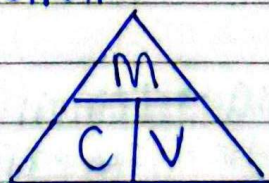
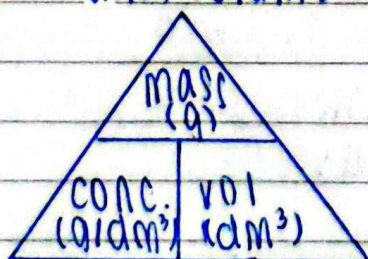
One mole of any gas occupies a volume of 24 dm^3 (24000 cm^3) at r.t.p.



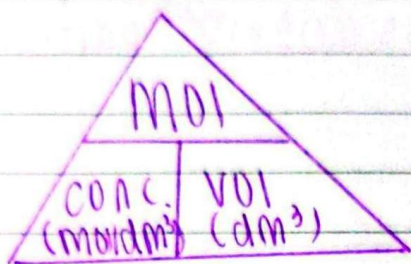
This formula applies to gas only.

Concentration of a solution in g/dm^3

Concentration of solution is the amount of solute dissolved per unit volume of solvent.



Concentration of a solution in mol/dm^3



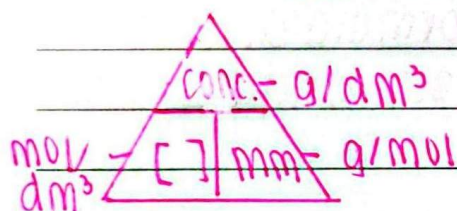
concentration of a solution:

$$\text{g/dm}^3 \longleftrightarrow \text{mol/dm}^3$$

$\div \text{mm}$

$$\text{g/dm}^3 \longleftrightarrow \text{mol/dm}^3$$

$\times \text{mm}$



Shorthand for mole calculation

$n_{\text{H}_2} \rightarrow$ number of moles of H_2

$N_{\text{H}_2} \rightarrow$ number of particles of H_2

$V_{\text{H}_2} \rightarrow$ volume of H_2

conc. of $\text{H}_2\text{SO}_4 \rightarrow$ concentration of H_2SO_4 in g/dm^3

$[\text{H}_2\text{SO}_4] \rightarrow$ concentration of H_2SO_4 in mol/dm^3

Empirical and Molecular Formulae

Empirical formula $\rightarrow$ simplest ratio of elements in a compound

Converting Empirical Formula to Molecular Formula

$$n = \frac{\text{Mr of compound}}{\text{Mr of emp. formula}}$$

where molecular formula $\rightarrow$ (empirical) $_n$
 $\rightarrow$ expand!

Calculate Empirical Formula from Mass / Percentage Mass
 If given percentage mass (%), write
 Assume mass of compound is 100g.

Draw table (3SF!)

Substance	(Element)	(Element)	(Element)	...
Mass/g				
Molar Mass in g/mol				
Number of moles /mol				
Mole ratio				
Simplest ratio				

Emp. formula:

Mole ratio: Divided no. of moles by smallest no. of moles

Calculate Molecular Formula from Mass / Percentage Mass
 and Mr of compound

Step 1: Empirical formula using table.

Step 2: Find n, then write (emp. formula)ⁿ and expand.

Stoichiometry - Mole Ratio

Step 1: Identify 'given' and 'to find'.

Step 2: Calculate no. of moles of 'given'.

Step 3: Compare mole ratio of 'given' and 'to find'

using balanced chemical equation.

(given) : (to find)

(ratio) : (ratio)

(no. of mol) : (no. of mol) × (ratio)

Volume Ratio of gases

Using Avogadro's Law, instead of using mole ratio,

use volume ratio if comparing volume. (ONLY FOR BOTH GASES)

Volumetric Analysis - titration (salt prep)

Usually given conc. and volume of one reactant and volume of other reactant, find conc. of other reactant.

Step 1: Convert to mol/dm^3 to find no. of mol of reactant.

Step 2: Mole ratio of the reactants; find no. of mol of other reactant

Step 3: Find conc. of other reactant in mol/dm^3

Step 4: If necessary, convert to g/dm^3 .

Limiting Reactants

Rule of thumb: Check for limiting reactants when the quantities of 2 or more reactants are given.

Answering LR qns:

Step 1: Find no. of mol for both reactants

Step 2: Mole Ratio to find LR:

(reactant 1) : (reactant 2)

$m : n$

$10 \text{ mol} : 10 \times \frac{n}{m} \text{ mol}$

Explanation (MUST WRITE IF QN ASKS TO IDENTIFY LR):

10 mol of (reactant 1) is needed to react with $(10 \times \frac{n}{m}) \text{ mol}$ of (reactant 2), but $k \text{ mol}$ of (reactant 2) is given.

$\therefore$ (reactant 1/ reactant 2) is the limiting reagent.

★ MUST CHECK LR FIRST BEFORE DOING MOLE RATIO W/ PRODUCTS ETC. (DO MOLE RATIO WITH LR NOT THE REACTANT IN EXCESS!)

Percentage yield and purity

Percentage yield = $\frac{\text{actual yield}}{\text{theoretical max. yield}} \times 100\%$

Percentage purity = $\frac{\text{mass of pure substance in sample}}{\text{total mass of impure sample}} \times 100\%$

Percentage Yield

Step 1: Balanced Chemical Equation

Step 2: Start from reactants:

Find no. of mol of given

Step 3: Mole ratio to find no. of mol of product

Step 4: Mass:

Theoretical Max. Mass of (product) =

Step 5: % yield

$$\% \text{ yield} = \frac{\text{given mass obtained}}{\text{theoretical max. mass}} \times 100\%$$

Percentage Purity

Step 1: Balanced Chemical Equation

Step 2: Start from products:

Find no. of mol of given

Step 3: Mole ratio to find no. of mol of reactant

Step 4: Mass of pure (reactant) =

Step 5: % purity

$$\% \text{ purity} = \frac{\text{mass of pure (reactant)}}{\text{Given mass of impure sample}} \times 100\%$$